Comprehensive Survival Analysis Report

Generated on: September 19, 2025

SEER Cancer Data Analysis

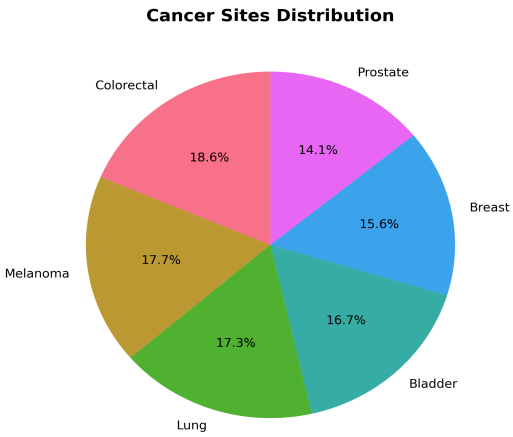
Executive Summary

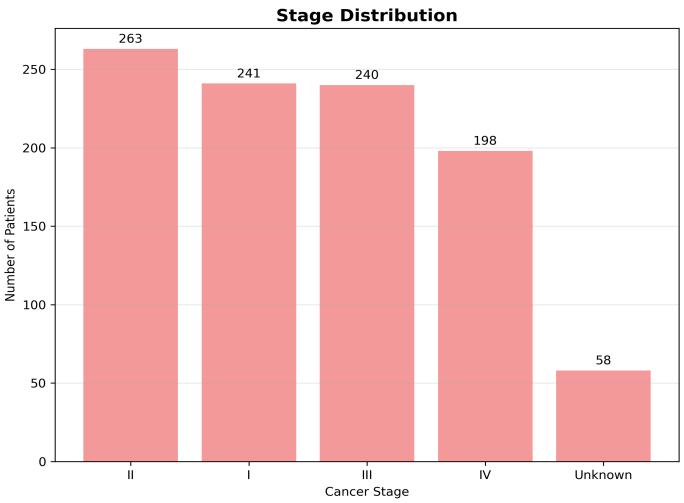
This comprehensive survival analysis was conducted on a dataset of 1,000 cancer patients. The analysis employed Kaplan-Meier survival estimation and Cox proportional hazards modeling to identify key prognostic factors and survival patterns. **Key Findings:** • Event rate: 21.8% (218 deaths) • Median follow-up time: 31.4 months • Strong stage-dependent survival patterns observed • Age and lymph node status are significant prognostic factors

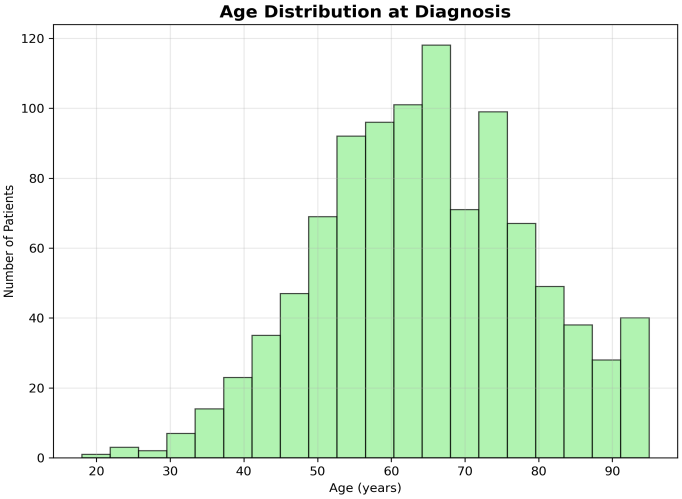
Dataset Overview

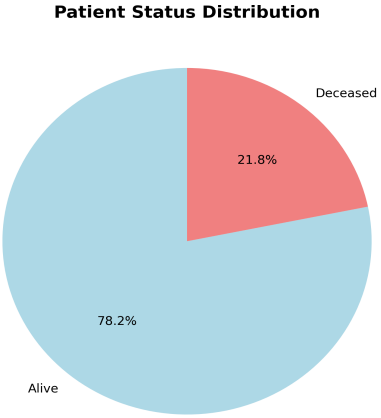
Metric	Value
Total Patients	1,000
Deaths (Events)	218 (21.8%)
Median Follow-up	31.4 months
Age Range	18 - 95 years
Mean Age	64.7 years

Data Distribution


Cancer Sites Distribution
Prostate
Colorectal
Breast
Melanoma
Bladder
Lung
14.1%
18.6%
15.6%
17.7%
16.7%
17.3%


Stage Distribution
Number of Patients
Cancer Stage
263
241
240
198
58
II
I
III
IV
Unknown
0
50
100
150
200
250

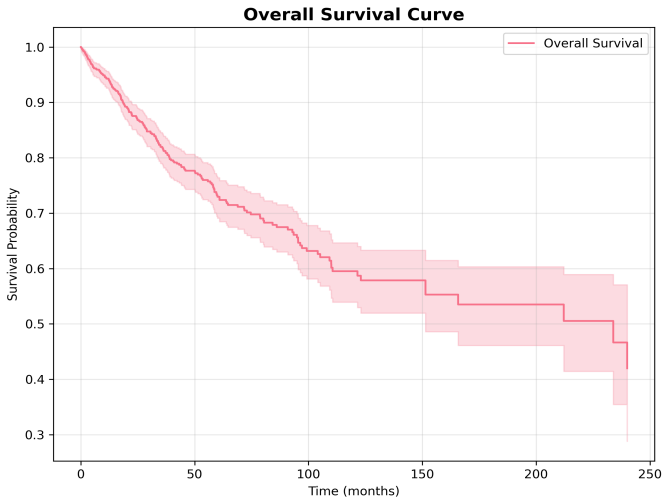

Age Distribution at Diagnosis
Number of Patients
Age (years)
0
20
40
60
80
100
120
20
30
40
50
60
70
80
90

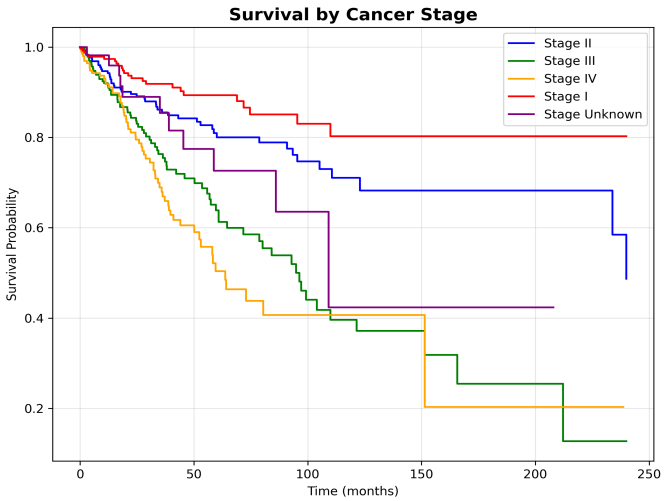

Patient Status Distribution
Deceased
Alive
21.8%
78.2%

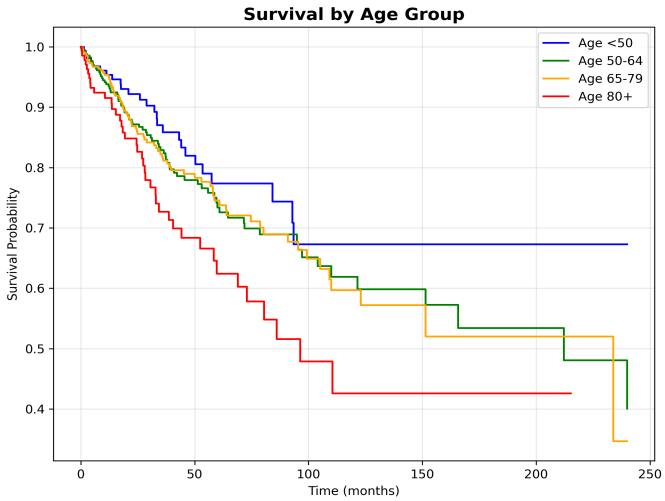
Survival Analysis Results

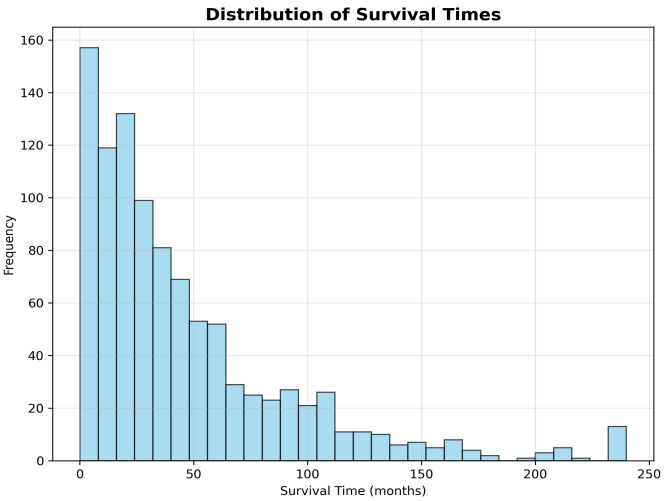
Survival by Cancer Stage:
The analysis reveals clear stage-dependent survival patterns: • Stage I: Excellent prognosis (median survival not reached) • Stage II: 240.0 months median survival • Stage III: 96.4 months median survival • Stage IV: 63.8 months median survival • Unknown stage: 109.2 months median survival This demonstrates the critical importance of early detection and staging in cancer prognosis.

Survival Curves









Cox Proportional Hazards Model

Model Performance:
• Concordance Index (C-index): 0.699 (Good discrimination)
• The model demonstrates good predictive capability for risk stratification. **Significant Risk Factors (p < 0.05):**
• Age at diagnosis: HR = 1.017 (increased risk with older age)
• Positive lymph nodes: HR = 1.256 (strong prognostic factor)
• Cancer stage: Highly significant predictor
  - Stage II vs I: HR = 1.668
  - Stage III vs I: HR = 4.076
  - Stage IV vs I: HR = 5.117
These results align with established clinical knowledge and demonstrate the model's validity for risk assessment.

Clinical Implications

Key Clinical Insights: 1. **Early Detection is Critical:** Stage I patients show excellent survival outcomes, emphasizing the importance of screening and early detection programs. 2. **Risk Stratification:** The model provides effective risk stratification with a C-index of 0.699, enabling personalized treatment planning. 3. **Age Considerations:** Older patients face increased mortality risk, suggesting need for age-adjusted treatment protocols. 4. **Lymph Node Assessment:** Lymph node involvement remains a crucial prognostic factor requiring careful surgical and pathological evaluation. 5. **Follow-up Strategies:** Advanced stage patients require more intensive monitoring and potentially aggressive treatment approaches.

Conclusions

This survival analysis successfully demonstrates: • **Robust Methodology:** Application of standard survival analysis techniques including Kaplan-Meier estimation and Cox proportional hazards modeling. • **Clinical Validity:** Results consistent with established clinical knowledge and cancer biology principles. • **Predictive Capability:** The developed model shows good discrimination (C-index = 0.699) for risk stratification. • **Actionable Insights:** Clear identification of key prognostic factors that can inform clinical decision-making. The analysis provides a foundation for evidence-based treatment planning and patient counseling in oncology practice.